



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Image Acquisition Process

Subtitle: Challenges, CCD vs CMOS Sensors, and System Model

ITA0515-COMPUTER VISION

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INTRODUCTION



- ❖ Image acquisition is the process of capturing images using a camera or sensor.
- ❖ It converts a real-world scene into a digital image.
- ❖ It is the first step in image processing.
- ❖ Used in medical imaging, surveillance, robotics, and smartphones.
- ❖ **Image:** Camera capturing an image.



OBJECTIVES



- ❖ Understand the image acquisition process.
- ❖ Identify real-world image capturing challenges.
- ❖ Learn the functions of image acquisition components.
- ❖ Compare CCD and CMOS sensors.
- ❖ Study the image acquisition system model.



IMAGE ACQUISITION PROCESS



Flow Diagram:

Object



Camera Lens



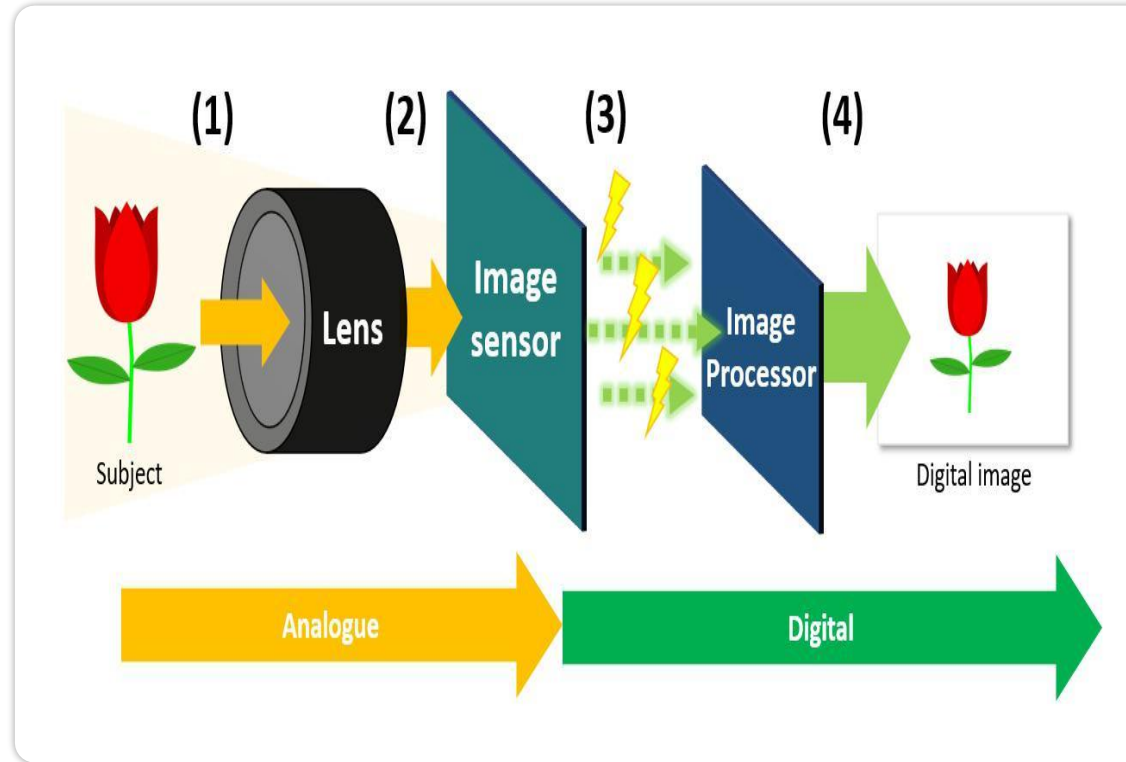
CCD / CMOS Sensor



ADC



Digital Image





Real-World Challenges



Challenges in Image Acquisition

- ❖ Low lighting
- ❖ Motion blur
- ❖ Image noise
- ❖ Camera shake
- ❖ Weather conditions
- ❖ Overexposure and underexposure

Effects:

- ❖ Poor image quality
- ❖ Loss of details
- ❖ Incorrect detection



CCD vs CMOS Sensors

CCD	CMOS
Better image quality	Faster processing
Low noise	Low power consumption
High power usage	Low cost
Expensive	Widely used in smartphones

Conclusion:

- CCD → Better quality
- CMOS → Faster and energy efficient



CONCLUSION

- ❖ Image acquisition converts real scenes into digital images.
- ❖ CCD and CMOS are the main image sensors.
- ❖ Image quality depends on lighting, motion, and sensor performance.
- ❖ CMOS sensors are commonly used because they are fast and affordable.

Thank
You

The image features the words "Thank You" written in a fluid, cursive script. The text is centered and surrounded by several stylized, hand-drawn shapes that resemble leaves or petals, arranged in a circular pattern around the central text. The entire design is rendered in black lines on a light gray background.